

Blackouts Preserving Rectangle Identifiability on Grid Points

A partial solution to Problem 001 of Kagey’s Open Problem Collection

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Abstract

We study Problem 001 of Kagey’s Open Problem Collection: black out as many points of an $n \times m$ grid as possible so that every possible rectangle drawn with its corners on grid points remains uniquely identifiable from its visible corners. We reduce the problem to a minimum hitting set of a difference hypergraph, solve the two-row case completely (the Strip Theorem: maximum blackout $m + 1$ with exactly $m \cdot 2^{m-1}$ optima for $m \geq 3$), prove that the maximum blackout is $m + 2$ on three rows for $m \geq 3$ and $m + 4$ on four rows for $m \geq 6$, and obtain the general upper bound $m + \lfloor n^2/4 \rfloor$ for $m \geq 3$. The structural step characterizes axis-aligned identification by the absence of four- and six-cycles in the black row–column incidence graph, and a separate geometric argument handles tilted rectangles. On the 5×5 grid the same characterization gives the value 10 by hand, since a bipartite graph on 10 vertices with no four- or six-cycle has at most 10 edges. We also report the computations, the uniqueness of the 5×5 optimum up to symmetry (by computer), and the remaining conjecture for larger grids.

1 Introduction

The problem is posed with some alterations chosen for clarity as the following: Suppose you, the player, are shown an $n \times m$ grid of vertices. Each vertex has two states, ‘white’ or ‘blackened out’. A vertex that is blackened out is unable to convey any information. The game master asks us to ‘black out’, or turn off, as many of the vertices as we like. Once we choose the ‘blackout’, which is simply the set of all vertices that are blackened out, the game master draws a rectangle with its corners on grid vertices. We cannot see the rectangle itself. We are only told which of its corners are white, so a corner on a blackened-out vertex tells us nothing. The rectangle’s sides are not necessarily parallel to the grid axes. We must be able to guess what the rectangle’s corners are from this information with complete confidence. The question posed by Kagey [1] is how many vertices can be blackened out while every rectangle the game master might draw remains distinctly identifiable.

The order of play is important here. Because the blackout is chosen by us before the rectangle is chosen, a blackout is valid only if *no* two rectangles are confusable under it. Writing $C(R)$ for the corner set of a rectangle and S for the blackout, what the player sees is $C(R) \setminus S$; the requirement is that this assignment be injective on the set of rectangles. Section 2 makes this precise, and the rest of the paper studies the resulting extremal quantity.

The Strip Theorem (Theorem 1) settles two rows, including the classification and number of optima. Theorems 3 and 4 settle the values on three rows and on four rows of width at least 6. The four-row grids of width 4 and 5 are settled by an exhaustive computer check (Corollary 2). The general row bound and the short-cycle characterization are Theorem 2 and Lemma 3. The 5×5

*Code and data: github.com/apemm/Kagey-Problems.

value is 10 (Theorem 5); the proof is a short counting argument on the same graph, plus a check of one construction. Optimum counts beyond strips and the 5×5 uniqueness statement remain computational. The higher-dimensional Excess-1 formula remains conjectural (Conjecture 1). The old guess that the kept set always has size $(n-1)(m-1)$ was first refuted at 5×5 and also fails on $4 \times m$ for every $m \geq 6$.

Every prediction was written in a ledger before the computation that tested it. The values were computed with an ILP formulation, a branch-and-bound oracle, and an exhaustive check straight from the definition, and Table 1 says which of these agree for each grid. The three- and four-row results have their own verifier, which enumerates the rectangles by their diagonals.

The paper is organized as follows. Section 2 fixes conventions and defines the presentation map. Section 3 proves the reduction and its consequences. Section 4 describes the pruning of the difference hypergraph. Section 5 proves the Strip Theorem and Section 6 proves the row bound and the results for three and four rows. Section 7 reports the computations and the verification. Section 8 proves the value at 5×5 and describes its unique optimum. Section 9 records the refuted and live conjectures and states open questions.

2 Conventions and the presentation map

Vertices of the $n \times m$ grid are denoted as pairs (i, j) with $1 \leq i \leq m$ (column) and $1 \leq j \leq n$ (row). The set of all vertices on the grid is V . The rectangle class consists of all rectangles with four corners on grid points, tilted rectangles included and degenerate (zero-area) configurations excluded; $C(R)$ denotes the corner set of a rectangle R . For a blackout $S \subseteq V$ the *presentation map* sends each rectangle to its visible corners, $P_S(R) = C(R) \setminus S$, and S is *valid* when P_S is injective over all rectangles. The empty presentation is an ordinary and accepted value of the map, that at most one rectangle may be fully hidden is a corollary of injectivity, not an axiom (Remark 1).

Remark 1. Suppose 2 distinct rectangles $R_1 \neq R_2$ with corner sets C_1 and C_2 are both fully hidden, that is, $C_1 \subseteq S$ and $C_2 \subseteq S$. Then $P_S(R_1) = \emptyset = P_S(R_2)$, and since $\emptyset$ is an ordinary value of the presentation map, injectivity gives $R_1 = R_2$, contradicting our supposition. Hence, at most 1 rectangle may be fully hidden under any valid blackout, with no separate convention required.

3 The reduction

Lemma 1 (difference-set characterization). *For any sets A, B, S : $A \setminus S = B \setminus S$ if and only if $A \triangle B \subseteq S$.*

Proof. ($\Rightarrow$) Assume $A \setminus S = B \setminus S$, and let $x \in A \triangle B$ be arbitrary. Exchanging A and B makes no difference since $\triangle$ is commutative and symmetric about A and B , so we may assume $x \in A \setminus B$. Suppose $x \notin S$. Then $x \in A \setminus S$, and since $A \setminus S = B \setminus S$ we get $x \in B \setminus S$, hence $x \in B$, contradicting $x \in A \setminus B$. Therefore $x \in S$. As x was arbitrary, $A \triangle B \subseteq S$.

($\Leftarrow$) Assume $A \triangle B \subseteq S$. Both the hypothesis and the conclusion are unchanged when A and B are exchanged, so it suffices to prove $A \setminus S \subseteq B \setminus S$. Let $x \in A \setminus S$ be arbitrary, so $x \in A$ and $x \notin S$. Suppose $x \notin B$. Then $x \in A \setminus B \subseteq A \triangle B \subseteq S$, contradicting $x \notin S$. Hence $x \in B$, and with $x \notin S$ we get $x \in B \setminus S$. As x was arbitrary, $A \setminus S \subseteq B \setminus S$; the reverse inclusion follows by symmetry, so $A \setminus S = B \setminus S$. $\square$

Corollary 1. *S is valid if and only if the kept set $K = V \setminus S$ intersects the difference set $C(R_1) \triangle C(R_2)$ for every pair of distinct rectangles. Hence a maximum blackout is the complement of a minimum hitting set of the difference hypergraph.*

Proof. By definition, S is valid if and only if $P_S(R_1) \neq P_S(R_2)$, i.e. $C(R_1) \setminus S \neq C(R_2) \setminus S$, for every pair of distinct rectangles. From Lemma 1, this holds if and only if $C(R_1) \triangle C(R_2) \not\subseteq S$, which is exactly the statement that at least 1 point of the difference set lies in K . $\square$

Two simple facts about rectangles are used throughout.

Lemma 2 (three corners suffice). *Three corners of a rectangle determine the rectangle. Consequently, if two rectangles have the same presentation under some blackout and that presentation has at least 3 points, the two rectangles are equal.*

Proof. Let p, q, r be 3 corners of a rectangle. Exactly one of them, say q , is adjacent to the other 2, and the angle at q in the triangle pqr is a right angle. A triangle has at most 1 right angle, so q is determined by the 3 points alone, and then the fourth corner is $p + r - q$. For the second claim, a presentation with at least 3 points contains 3 corners of each rectangle, and these are the same 3 points, so the corner sets agree. $\square$

In particular, two rectangles can only be confused if they have the same number of visible corners, and that number is at most 2. So when we check a blackout, only rectangles with at least 2 black corners need attention.

4 Preprocessing

Before any computation we shrink the difference hypergraph. Think of it as a table with one line for each pair of distinct rectangles, listing their difference set; a kept set K is valid when it meets every line. To avoid a clash with the rows of the grid, we call these lines *constraints*. The first step is deduplication: identical constraints say the same thing, so we keep one copy. The second step is a containment rule: if $D_1 \subsetneq D_2$ are constraints, any K meeting D_1 automatically meets D_2 , so D_2 can be deleted without changing which K are valid. Finally, no constraint is empty: $C_1 \triangle C_2 = \emptyset$ would mean $C_1 = C_2$, and distinct rectangles have distinct corner sets.

After these steps the hypergraph is replaced by the antichain of its minimal constraints, and a set K meets every original constraint if and only if it meets every minimal one. All of the following computations in Section 7 run on the pruned version of the hypergraph.

5 The Strip Theorem

Theorem 1 (Strip Theorem). *For a $2 \times m$ grid with $m \geq 3$, every maximum valid blackout has size $m + 1$, there are exactly $m \cdot 2^{m-1}$ of them, and they are characterized as the complements of kept sets containing one survivor per column and missing exactly one column. For $m = 2$ the condition is vacuous and every blackout is valid.*

Proof. Define K as the visible set, where $V \setminus S = K$, and let K_0 denote the minimum possible size of K over valid blackouts. For a $2 \times m$ grid, define the set T to be the set of all columns with at least 1 of their 2 points in K . For $m = 2$ there is exactly one rectangle, so injectivity is vacuous and every blackout is valid; assume $m \geq 3$. Suppose S is a maximum valid blackout.

On a $2 \times m$ strip every rectangle is axis-aligned, since a tilted rectangle's corners would occupy at least 3 distinct rows and the strip has only 2. We can represent any possible rectangle R as $R_{i,j}$, where i is the column containing the left corners of R and j is the column containing the right corners, $1 \leq i < j \leq m$.

If two distinct rectangles share exactly one column, say $R_{z,p}$ and $R_{z,q}$, the shared column's corners overlap, and $C(R_{z,p}) \triangle C(R_{z,q})$ is exactly the 4 points of columns p and q ; we call this the pair constraint on $\{p, q\}$. If they share no column, the difference set is all 8 points of 4 distinct columns, which contains a pair constraint and is deleted in preprocessing (Section 4). Every pair $\{p, q\}$ arises this way: two rectangles sharing a column span at least 3 distinct columns between them, and $m \geq 3$ guarantees a third column z from which to build $R_{z,p}$ and $R_{z,q}$. Hence, the pruned hypergraph consists of exactly the $\binom{m}{2}$ pair constraints.

Then, from Lemma 1, a valid visible set K is one that intersects every pair constraint. Conditioning on T , this constrains T to contain every column except at most one: if two distinct columns p and q are excluded, the pair constraint on $\{p, q\}$ contains no point of K , and K would not intersect every pair constraint. Each column of T contributes at least 1 point, so $|K| \geq |T| \geq m - 1$, and hence $|S| \leq 2m - (m - 1) = m + 1$. (The construction below shows that $m - 1$ is attained, so $K_0 = m - 1$.) If no column is fully blacked out, T would contain every column and K would need m points, so $|S| \leq m$. If any column of T kept both of its points, $|K| \geq m$ as well. Hence, at the maximum, exactly one column is excluded from T , and in the other $m - 1$ columns, exactly 1 of the 2 points in each column is blacked out.

Every such construction is valid: any pair of columns includes at least one column of T , whose surviving point meets the pair constraint. There are m ways to choose which column to exclude from T and 2^{m-1} ways to choose which 1 of the 2 points to black out in the remaining $m - 1$ columns. Hence, $m \cdot 2^{m-1}$ such distinct maximum blackouts exist, characterized as in the statement. $\square$

6 Three and four rows

In this section coordinates start at zero: columns are $0, \dots, m - 1$ and rows are $0, \dots, n - 1$.

The plan is as follows. Most rectangles are axis-aligned, and for those the question turns into a question about a graph: Lemma 3 says exactly which blackouts keep the axis-aligned rectangles apart. A classical extremal bound on that graph gives the upper bound (Theorem 2). On three and four rows there are only a few kinds of tilted rectangles (Lemma 4), so for the lower bound we can write down a blackout and check the tilted rectangles by hand.

To see where the graph comes from, think of a black point (x, y) as a link between row y and column x . An axis-aligned rectangle uses 2 rows and 2 columns, and its 4 corners are the 4 possible links between them. If all 4 are black, the 4 links form a closed loop of length 4, and the next lemma shows that such loops, and loops of length 6, are exactly what causes confusion.

Lemma 3 (axis-aligned identification and short cycles). *Suppose $m \geq 3$. Form a bipartite graph G_S whose vertices are the rows and columns, with an edge between row y and column x exactly when $(x, y) \in S$. The axis-aligned rectangles have distinct presentations if and only if G_S has neither a cycle of length four nor a cycle of length six.*

Proof. If rows a, b are black in both columns x, z , choose a third column w . The rectangles on these rows and column pairs $\{x, w\}$ and $\{z, w\}$ have the same presentation. Thus a four-cycle is forbidden. If a six-cycle has successive vertices a, x, b, y, c, z, a , the rectangles with row/column pairs $(\{a, b\}, \{x, y\})$ and $(\{a, c\}, \{y, z\})$ have every corner black except possibly their common corner (y, a) . Their presentations coincide.

Conversely, assume both cycles are absent. Let $R_1 \neq R_2$ be axis-aligned rectangles with the same presentation P ; we derive a contradiction by cases on $|P|$. If $|P| \geq 3$, Lemma 2 gives $R_1 = R_2$. If $|P| = 0$, then R_1 is entirely black, and its 4 corners are the 4 edges of a four-cycle. Suppose $|P| = 2$. If the 2 visible points are in different rows and different columns, they are opposite corners

of both rectangles, and 2 opposite corners determine an axis-aligned rectangle. If they lie in one row a , at columns x and y , then R_1 and R_2 use rows $\{a, b\}$ and $\{a, c\}$ with $b \neq c$, and the 4 points $(x, b), (y, b), (x, c), (y, c)$ are black, a four-cycle. The case of one column is the same with rows and columns exchanged. Finally, suppose $|P| = 1$, so the common presentation is the single corner (x, a) . Write the other row and column of the two rectangles as (b, y) and (c, z) . If $b = c$ or $y = z$, their black corners contain a four-cycle. Otherwise the six black edges form the cycle a, y, b, x, c, z, a . Every possibility contradicts the assumption. $\square$

Theorem 2 (general row bound). *For $m \geq 3$, every valid blackout on an $n \times m$ grid satisfies*

$$|S| \leq m + \left\lfloor \frac{n^2}{4} \right\rfloor.$$

If also $n \geq 3$, the transposed bound holds as well.

Proof. For each nonempty column let B_x be its set of black rows, and choose any tree on B_x (no edges if $|B_x| = 1$). Superimpose these trees into a graph H on the n row vertices. It is simple: repeating an edge would mean two columns share two black rows, a four-cycle in G_S . It is triangle-free. A triangle cannot lie within one tree; if two edges come from one column and the third from another, those columns share two rows; if the three edges come from three columns, they give a six-cycle in G_S . All cases are forbidden by Lemma 3.

Recall that a triangle-free graph on n vertices with $e > 0$ edges satisfies $d(u) + d(v) \leq n$ for every edge uv . Summing over edges and using Cauchy–Schwarz gives

$$\frac{4e^2}{n} \leq \sum_v d(v)^2 \leq ne,$$

so $e \leq \lfloor n^2/4 \rfloor$, the usual Mantel bound. If $t \leq m$ columns are nonempty, the construction gives $e = \sum_{B_x \neq \emptyset} (|B_x| - 1) = |S| - t$. Therefore $|S| = t + e \leq m + \lfloor n^2/4 \rfloor$. $\square$

Lemma 4 (tilted rectangles on narrow grids). *Every tilted rectangle on three rows is a translate of*

$$D(x, y) = \{(x, y+1), (x+1, y), (x+2, y+1), (x+1, y+2)\}, \quad y = 0.$$

On four rows, these diamonds have $y = 0, 1$, and the remaining tilted rectangles are the following four types, for $0 \leq x \leq m-4$:

$$\begin{aligned} A_x &= \{(x, 2), (x+1, 3), (x+2, 0), (x+3, 1)\}, \\ B_x &= \{(x, 2), (x+2, 3), (x+1, 0), (x+3, 1)\}, \\ C_x &= \{(x, 1), (x+1, 3), (x+2, 0), (x+3, 2)\}, \\ E_x &= \{(x, 1), (x+2, 3), (x+1, 0), (x+3, 2)\}. \end{aligned}$$

Proof. A tilted rectangle has a unique leftmost corner. Indeed, suppose 2 corners have the smallest x -coordinate x_0 . They are not adjacent, since no side is vertical, so they are opposite and the midpoint of the rectangle has x -coordinate x_0 . The other 2 corners have the same midpoint, so their x -coordinates add up to $2x_0$; both are at least x_0 , so both equal x_0 , and all 4 corners lie on one vertical line, which is impossible. Both sides at the leftmost corner point strictly to the right, and since they are orthogonal one points up and one points down. At the leftmost corner, write the side vectors as (a, b) and $(c, -d)$, where a, b, c, d are positive integers. Orthogonality says $ac = bd$, and the vertical span is $b + d$. If this span is at most two, all four integers are one. If it is three, $\{b, d\} = \{1, 2\}$ and $\{a, c\} = \{1, 2\}$, giving the four listed cases. These exhaust the possibilities. $\square$

Theorem 3 (Three-Row Theorem). *For every $m \geq 3$, the maximum valid blackout on a $3 \times m$ grid is $m + 2$.*

Proof. The upper bound is Theorem 2. For the lower bound take

$$S = \{(x, 0) : 0 \leq x < m\} \cup \{(0, 1), (0, 2)\}.$$

Its incidence graph is a tree, so its axis-aligned presentations are distinct. By Lemma 4, each tilted rectangle is a diamond $D(x, 0)$. For $x > 0$ it has only one black corner, hence at least three visible corners and cannot be confused with any other rectangle. For $x = 0$ its presentation is $\{(2, 1), (1, 2)\}$. The unique axis-aligned rectangle with these opposite corners also contains the white points $(1, 1), (2, 2)$, so cannot have this presentation. There is no second tilted rectangle with at most two visible corners. Thus S is valid and has $m + 2$ points. $\square$

Theorem 4 (Four-Row Theorem). *For every $m \geq 6$, the maximum valid blackout on a $4 \times m$ grid is $m + 4$.*

Proof. The upper bound is Theorem 2. For the matching construction set

$$S = \{(x, 0) : 1 \leq x \leq m - 2\} \\ \cup \{(m - 2, 1), (m - 1, 1), (0, 2), (1, 2), (0, 3), (m - 1, 3)\}.$$

Its size is $m + 4$. Its columns with two black points are $0, 1, m - 2, m - 1$, joining row pairs $\{2, 3\}, \{0, 2\}, \{0, 1\}, \{1, 3\}$, respectively. Thus G_S is an eight-cycle with pendant edges, and all axis-aligned presentations are distinct.

It remains to check tilted rectangles. Any rectangle with at least three visible corners is uniquely determined, so only rectangles with at least two black corners need attention. All diamonds $D(x, 1)$, and all C_x, E_x , have at most one black corner. Put $L = \{0, 1\}$ and $R = \{m - 5, m - 4\}$. The remaining presentations with at most two visible corners are exactly those in the table. In the last three rows $F = A$ or B , with $s = 1$ or 2 , respectively.

rectangle	range	visible corners
$D(0, 0)$		$\{(0, 1), (2, 1)\}$
$D(x, 0)$	$x = m - 4, m - 3$	$\{(x, 1), (x + 1, 2)\}$
F_x	$x \in L \setminus R$	$\{(x + s, 3), (x + 3, 1)\}$
F_x	$x \in R \setminus L$	$\{(x, 2), (x + s, 3)\}$
F_x	$x \in L \cap R$	$\{(x + s, 3)\}$

The table follows by checking membership in S . Its two-point presentations are distinct: their row pairs and horizontal separations distinguish the types, and within a type the coordinates determine x . The only one-point presentations occur at $m = 6, x = 1$, and are $(2, 3)$ and $(3, 3)$, which are distinct.

None equals an axis-aligned presentation. For $\{(0, 1), (2, 1)\}$ no other row is black in both columns $0, 2$. For $\{(x, 1), (x + 1, 2)\}$ the other axis-aligned corner $(x, 2)$ is white, since $x \geq 2$. In the $L \setminus R$ case, $(x + s, 1)$ is white, since $x + s \leq 3 < m - 2$. In the $R \setminus L$ case, $(x, 3)$ is white, since $0 < x < m - 1$. Finally, when $m = 6$, an axis-aligned rectangle with sole visible corner $(z, 3)$, $z = 2$ or 3 , would need another column $c = 0$ or 5 and another row black in both columns z, c . No such row exists: row zero is white at $0, 5$, and rows one and two are white at z . Hence S is valid. $\square$

Lemma 5 (four-row upper-bound equality cases). *If $m \geq 4$ and a valid blackout has $m + 4$ black points on four rows, then every column is nonempty, exactly four columns have two black points, the others have one, and the four double columns form a four-cycle on the row vertices.*

Proof. Equality in Theorem 2 requires every column nonempty and total excess $\sum_x (|B_x| - 1) = 4$. A column containing all four rows leaves all other columns with at most one black point, giving excess three. If a column contains three rows, any other nonsingleton column must pair the fourth row with one of those three. There can be at most one such column: two with the same row pair give a four-cycle in G_S , and two with different row pairs give a six-cycle. The total excess is then at most $2 + 1 = 3$. Thus all nonsingleton columns have size two. They give a simple triangle-free graph with four vertices and four edges (the graph H in the proof of Theorem 2). Such a graph is a four-cycle: its complement has 2 edges, and if they shared a vertex v , the other 3 vertices would form a triangle; so the 2 missing edges are disjoint, and what remains is a four-cycle. $\square$

Corollary 2 (finite boundary cases, computer-assisted). *The maximum blackout is 7 on 4×4 and 8 on 4×5 .*

Proof. The upper bound leaves only a possible value $m + 4$ to exclude. By Lemma 5, candidates are generated by choosing one of the three four-cycles on the row labels, four columns, a bijection from their four row-pair edges to those columns, and a row for each remaining singleton column. This is the complete list of

$$3 \cdot 4! \binom{m}{4} 4^{m-4}$$

candidates: 72 when $m = 4$ and 1440 when $m = 5$. The standard-library script `verify_narrow.py` checks every candidate directly against all rectangles, generated independently from equal-length diagonals with a common midpoint. Every candidate has a repeated presentation. The same script checks that the blackout consisting of row zero and column zero is valid in these two grids, attaining $m + 3$. $\square$

The verifier also checks Lemma 3 on all 74,304 blackouts of five small grids, compares the lists of tilted rectangles in Lemma 4 with the diagonal enumeration, and checks the constructions for widths 3, ..., 40, 64, 100, 128 where they apply. These checks are only a test of the proofs above; the proofs do not depend on them.

7 Computations and verification

The last column says which methods agree on each value. The values for three and four rows also follow from Section 6, but the counts of optima are only computational. Rectangle counts match the literature: $n \times n$ counts 1, 10, 44, 130 agree with OEIS A085582, and $3 \times m$ counts with OEIS A289832 (indexing note in the bibliography).

8 The 5×5 grid

The 5×5 grid is the first square where the old guess of $n + m - 1$ black points fails, and it fails by exactly 1. It is also the smallest grid where both dimensions are at least 5, the case Conjecture 1 is about. The row bound (Theorem 2) only gives $5 + \lfloor 25/4 \rfloor = 11$ here. In this section we show that the true value is 10, and we describe the optimum.

The idea is to look at the blackout as a graph, as in Lemma 3: one vertex for each row, one for each column, and an edge for each black point. On a 5×5 grid this graph has only 10 vertices. Lemma 3 forbids cycles of length 4 and 6, and the graph is bipartite, so every cycle has length at least 8. On 10 vertices there is room for 1 such cycle but not for 2 that overlap, and 2 that do not

grid	rects	blackout	kept	optima	verification
2×2	1	4	0	1	three-way
2×3	3	4	2	12	three-way
2×4	6	5	3	32	three-way
2×5	10	6	4	80	three-way; pre-registered
2×6	15	7	5	192	engine + oracle
2×7	21	8	6	448	engine + oracle
3×3	10	5	4	53	three-way
3×4	20	6	6	224	three-way
3×5	33	7	8	892	three-way
3×6	49	8	10	3420	three-way
4×4	44	7	9	316	three-way
4×5	74	8	12	1810	three-way
3×7	68	9	12	12704	engine + brute
3×8	90	10	14	45864	engine + brute
3×9	115	11	16	161992	engine + brute
3×10	143	12	18	—	engine only (value)
4×6	110	10	14	26	engine + brute
4×7	152	11	17	292	engine + brute
4×8	200	12	20	—	engine only (value)
5×5	130	10	15	8	engine + brute
5×6	198	11	19	—	engine only (value)
5×7	276	12	23	—	engine only (value)
6×6	313	12	24	—	engine only (value)

Table 1: Exact values. “Three-way” = engine, oracle, and definitional brute agree; “engine + brute” = the ILP engine and the exhaustive definitional check agree; “engine only” = the exhaustive check is infeasible at that size.

overlap need at least 16 vertices. So the graph has at most as many edges as vertices. The next lemma makes this precise.

Lemma 6 (few edges without short cycles). *Let G be a bipartite graph with $v \leq 10$ vertices and no cycle of length 4 or 6. Then G has at most v edges. If it has exactly v edges, then every connected component of G contains exactly 1 cycle, so G is connected and its only cycle has length 8 or 10.*

Proof. Since G is bipartite, all of its cycles have even length, so every cycle has length at least 8.

Step 1: no component has more edges than vertices. Suppose some component G' does. Delete vertices of degree 1 from G' one at a time, each together with its edge, until none is left. Each deletion removes 1 vertex and 1 edge, so the result H still has more edges than vertices, and H is nonempty with every degree at least 2. If every degree in H were exactly 2, H would have as many edges as vertices. So some vertex x of H has degree at least 3.

Let D_i be the set of vertices of H at distance i from x . Two different paths from x to the same vertex, each of length at most 3, would together contain a cycle of length at most 6, which is impossible. So for $i = 1, 2, 3$ every vertex of D_i has exactly 1 neighbour in D_{i-1} , its *parent*. It has no neighbour in D_i , because D_i lies on one side of the bipartition. Since its degree is at least 2, it therefore has a neighbour in D_{i+1} . Two different vertices of D_i ($i \leq 2$) cannot share such a neighbour, since that neighbour would then have 2 parents. Hence $|D_1| \geq 3$, $|D_2| \geq |D_1| \geq 3$,

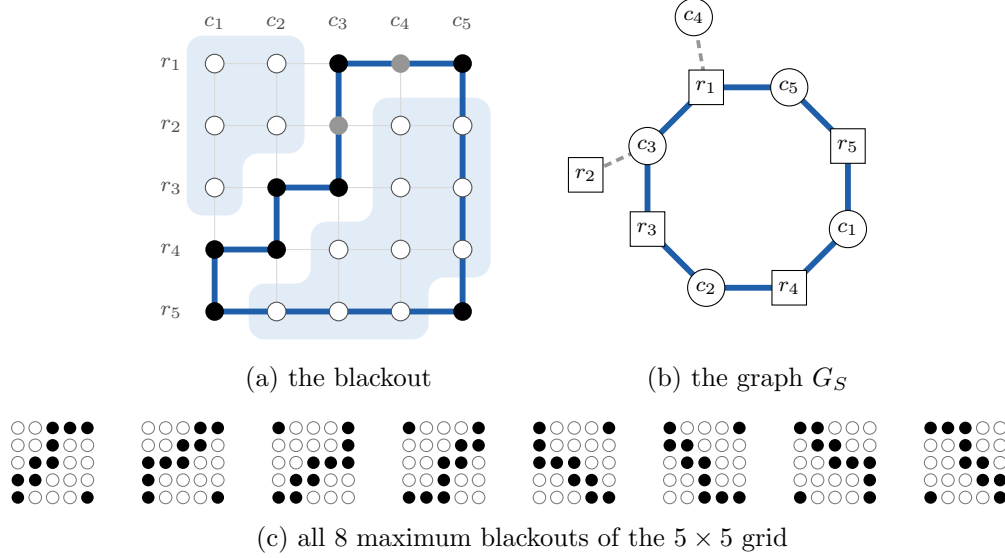


Figure 1: The 5×5 optimum. (a) The 10 blacked-out points (filled) and the 15 kept points (open), which form the 2 shaded blocks of 5 and 10. The bold closed path turns only at black points and runs alternately along rows and columns; its 8 corners are the 8-cycle of (b). The 2 grey black points are the 2 pendant edges of (b). (b) The same blackout as a graph: one vertex for each row and each column, and an edge $r_j c_i$ for each black point (i, j) . It is an 8-cycle with 2 pendant edges. A valid blackout can have no 4-cycle and no 6-cycle (Lemma 3), so 8 is the shortest cycle allowed. (c) The 8 maximum blackouts. They are the images of (a) under the 8 symmetries of the square, and no 2 of them coincide.

$|D_3| \geq |D_2| \geq 3$ and $|D_4| \geq 1$, and H has at least $1 + 3 + 3 + 3 + 1 = 11$ vertices, more than $v \leq 10$. This contradiction proves Step 1.

Step 2: the equality case. By Step 1, every component has at most as many edges as vertices, so the total number of edges is at most v . If it equals v , every component has exactly as many edges as vertices, and a connected graph with as many edges as vertices contains exactly 1 cycle. Two components would then hold 2 vertex-disjoint cycles of length at least 8, which needs $16 > 10$ vertices. So G is connected, and its cycle is even with length between 8 and $v \leq 10$. $\square$

Theorem 5 (the 5×5 grid). *The maximum blackout on the 5×5 grid is 10. For every maximum blackout S , the graph G_S is connected and contains exactly 1 cycle, of length 8 or 10. By computer, there are exactly 8 maximum blackouts. They form a single orbit of the symmetry group D_4 of the square, with trivial stabilizer, and each of their graphs is an 8-cycle with 2 pendant edges (Figure 1).*

Proof. Upper bound. Let S be a valid blackout. By Lemma 3 (which applies since $m = 5 \geq 3$), the graph G_S has no cycle of length 4 or 6. It is bipartite with $5 + 5 = 10$ vertices and $|S|$ edges, so Lemma 6 gives $|S| \leq 10$, and if $|S| = 10$ it gives the structure stated.

Lower bound. The blackout of Figure 1(a) has 10 points. Its graph is an 8-cycle with 2 pendant edges, so by Lemma 3 no 2 axis-aligned rectangles are confused. The remaining $130 - 100 = 30$ rectangles are tilted, and `verify_narrow.py` checks all 130 presentations directly and finds them distinct.

The count. By the upper bound, a maximum blackout has a graph with exactly 10 edges and no 4- or 6-cycle. The script lists all 44,640 such sets of 10 points (a search that only adds a point when no short cycle appears) and checks each against all 130 rectangles. Exactly 8 are valid, and

each has degrees 3, 3 at 2 vertices, 1, 1 at 2 others and 2 elsewhere, so its graph is an 8-cycle with 2 pendant edges. Applying the 8 symmetries of the square to the blackout in Figure 1(a) gives 8 different blackouts, and these are the 8 found. The count of 8 was found earlier, independently, by the ILP engine and by an exhaustive check of all blackouts from the definition (Table 1). $\square$

In words, the extremal blackout is a staircase of 9 points running from the top right corner to the bottom left, the old $n + m - 1$ pattern, plus the corner $(5, 5)$. The staircase alone has a graph that is a tree. The extra corner closes a cycle of length 8, the shortest the lemma allows. Of the 1,440 sets of 10 points whose graph is a single 10-cycle (2 black points in every row and column), none is valid; there the tilted rectangles are what goes wrong. The kept set is 2 blocks of 5 and 10 points on either side of the staircase, and the extremal blackout is unique up to symmetry while having no symmetry of its own.

9 Refuted and live conjectures

conjecture	status	killed by
kept = $(n - 1)(m - 1)$	refuted (pre-registered)	5×5
H_{sum} : blackout = $n + m - 1 + \lceil n + m \geq 10 \rceil$	refuted (pre-registered)	3×7
H_{area} : blackout = $n + m - 1 + \lceil nm \geq 24 \rceil$	refuted (pre-registered)	3×8
Four-Row: kept($4, m$) = $\lceil 5m/2 \rceil - 1$	refuted (pre-registered)	4×8

The Three-Row conjecture, which we checked for $3 \leq m \leq 10$ before proving it, is now Theorem 3. The four-row case of Excess-1 is Theorem 4, with the boundary cases in Corollary 2. We have not classified or counted the optima beyond two rows.

Conjecture 1 (Excess-1). $\text{blackout}(n, m) = n + m$ for $\min(n, m) \geq 4$, *excepting* 4×4 and 4×5 up to transposition, where it is $n + m - 1$.

We checked this on seven cells (4×6 through 6×6), including the prediction of 12 at 6×6 , which was written down before the 1063-second computation. The four-row cases are now proved, and so is 5×5 (Theorem 5). The other cases where both dimensions are at least 5 are open. There the row bound does not imply Excess-1, and neither does Lemma 6, which needs $n + m \leq 10$.

Beyond the remaining conjecture, three questions strike us as the natural next ones: why the number of optima collapses at 4×6 (26, against neighbors in the hundreds and thousands); whether the excess ever grows past 1, with 5×8 and 6×7 registered at 13 as the next kill attempts; and whether the 5×5 uniqueness admits a structural proof rather than a computational one. Theorem 5 reduces that question to 10-point blackouts whose graph is connected with a single cycle of length 8 or 10.

Data and code availability

The code, tables, and ledger are at <https://github.com/apemm/Kagey-Problems>. The two four-row boundary cases are computer-assisted.

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The Lean code for this paper was generated by AI. AI was also used to revise some of the writing and to help cross-verify the code.

References

- [1] P. Kagey, Problem 001, Open Problem Collection, peterkagey.com/problems/001/ (accessed August 2026).
- [2] OEIS Foundation, Sequence A085582: number of rectangles (orthogonal or not) with corners on an $n \times n$ grid of points. <https://oeis.org/A085582>. Verified against Table 1: 1, 10, 44, 130 at $n = 2..5$.
- [3] OEIS Foundation, Sequence A289832: table of rectangle counts on rectangular grids; diagonal equals A085582. <https://oeis.org/A289832>. Indexing: $T(n, k)$ counts rectangles on an $(n+1) \times (k+1)$ point grid, so an $r \times c$ point grid appears as $T(c-1, r-1)$; e.g. our 3×7 count is $T(6, 2) = 68$.